AWS EKS Architecture Guide

Amazon EKS

Amazon Elastic Kubernetes Service is a managed Kubernetes service on AWS. AWS manages the Kubernetes control plane infrastructure while customers manage workloads, networking choices, node capacity, and application configuration.

Worker Nodes

EKS workloads can run on managed node groups, self-managed nodes, or AWS Fargate. Managed node groups simplify lifecycle operations for EC2 worker nodes.

Networking

EKS commonly uses the Amazon VPC CNI plugin so Pods receive IP addresses from the VPC. Security groups and network policies can be used to control network access.

Load Balancing

AWS Load Balancer Controller can create AWS Application Load Balancers or Network Load Balancers for Kubernetes services and ingress resources. This connects Kubernetes traffic management with AWS load-balancing services.

Security

IAM roles can provide AWS permissions to workloads through mechanisms such as EKS Pod Identity or IAM Roles for Service Accounts. Least-privilege permissions should be used for production workloads.